Equilibrium in 2D

Exam code: 9MA0

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Equilibrium in 2D

How do I model particles in two dimensions?

- In **two dimensions**, a particle may be considered as existing and being able to move around two-dimensional space rather than in a single straight line (1D)
  - This 2D space is called a **plane** – so you may see the ***Oxy* plane** mentioned
- The directions of the two dimensions are chosen so that they are perpendicular to each other
- The two directions are usually
  - **horizontal** and **vertical**, or
  - **parallel** and **perpendicular** to an inclined plane

What is the resultant force and an unbalanced force in 2D?

- The **resultant force** in 2D is the **sum of forces** acting on a particle but this will now need to be considered in two parts – one for each direction
- An **unbalanced force** is a force acting on a particle where one or both parts are **not** cancelled by another force acting in the opposite direction
- So a **non-zero** resultant force will be non-zero in at **least one** of the two dimensions, is therefore **unbalanced** and the particle will **accelerate**

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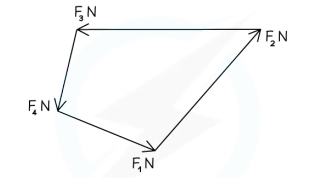
Ask questions and get help with exactly what you're revising.
Got it

resultant force acting on it is **zero**
    - For example, if the two dimensions involved are the **horizontal** and **vertical** directions, a particle will be in equilibrium if any forces acting **left** are **balanced** by any forces acting **right** and any forces acting **up** are **balanced** by any forces acting **down**

- When a particle has **several** forces (at least 3 – have a think as to why!) acting on it – and it is in **equilibrium** – the forces can be drawn "nose-to-tail" such that they form a polygon



A SYSTEM OF FOUR FORCES ACTING ON A PARTICLE IN EQUILIBRIUM

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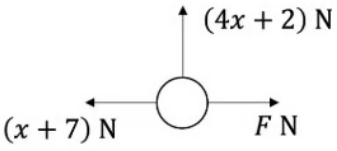
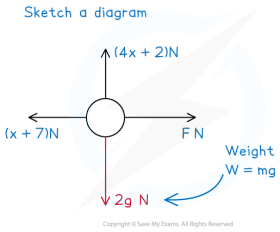
Worked Example

A particle of mass 2 kg is suspended in the air by three strings, one vertical and two horizontal. The particle is in equilibrium and the magnitudes of the tensions in the strings are illustrated in the diagram below.



(a) Write down the resultant forces acting on the particle in both the horizontal and vertical directions.

Answer:



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a) Particle is in equilibrium, so the resultant force is zero in both directions

∴ Resultant force in horizontal direction is 0 N.
Resultant force in vertical direction is 0 N.

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(b) Find the value of F.

Vertically: $(\uparrow)\ (4x + 2) - 2g = 0$

Solving the second equation

$4x = 2g - 2$

$x = \frac{g - 1}{2}$

Using $g = 9.8$ $x = \frac{8.8}{2} = 4.4$

Substitue $x = 4.4$ into the first equation

$F - (4.4 + 7) = 0$

$F = 11.4$

$\therefore F = 11.4\ N$

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Examiner Tips and Tricks

- F_{net} is often used for **resultant force** in equations.
- Sketching diagrams, or adding to any given in a question, can help you to understand a problem and pick up some marks.

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